

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 1.3: Quadratic Expressions and Equations — Teacher Reference (Pre-filled)

SUMMARY TABLE: CORE MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 1.3: Quadratic Expressions and Equations
Driving Question	DRIVING QUESTION: How can we use quadratic expressions and equations to find the dimensions of a plot when its fence length and area are both fixed, and why do two different plots sometimes have the same area?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Forming Quadratic Expressions from Areas: Mama Wanjiru's 40 m Fence	What we observed: With 40 m of wire, length + width = 20 m. Widths of 8 m and 12 m both give 96 m ² (the same 8 m by 12 m plot turned round), a width of 10 m gives the largest area of 100 m ² , and widths of 20 m or more do not give a plot.	What we learned: If the width is x m then the length is (20 – x) m and the area is $x(20 - x) = 20x - x^2$. Expanding $(x + 3)(x + 5)$ gives $x^2 + 8x + 15$. An expression whose highest power is 2 is a quadratic expression.	What we can explain so far: The x^2 term appears because multiplying a width x by a length that also contains x makes an x by x square region in the area model, so the area is a quadratic expression. We cannot yet explain why areas repeat in pairs, why the maximum is at 10 m, or how to find widths for a target area without trial and error; these questions go on the DQB.
Lesson 2 Deriving the Squares Identities from Area: Why $12^2 = (10 + 2)^2$ and $8^2 = (10 - 2)^2$	We built a 12 cm square from a 10 × 10 square, two 10 × 2 strips and a 2 × 2 square, and found its area is 144. We cut a 2 cm border from a 10 cm square and left an 8 cm square of area 64. The 10 cm square has area 100.	An identity is true for all values. $(a + b)^2 = a^2 + 2ab + b^2$, and $(a - b)^2 = a^2 - 2ab + b^2$. The middle term 2ab comes from two strips, so $(a + b)^2$ is not $a^2 + b^2$. In $(a - b)^2$, the b^2 is added back because the corner was cut away twice.	The squares $12^2 = (10 + 2)^2 = 144$, $8^2 = (10 - 2)^2 = 64$ and $10^2 = 100$ agree with the identities. The 8 m and 12 m widths are the same plot turned round. Model v1 shows area pictures for expansions, and it does not yet explain why 100 m ² is the largest area or how to find both widths for a target area.
Lesson 3 Cutting a Small Square from a Large Square: Difference of Two Squares	We cut a small square from the corner of a large square and rearranged the L-shape into a rectangle. The rectangle had sides (a – b) and (a + b) and the same area as the L-shape, for example 10 cm and 4 cm gave $6 \times 14 = 84$ cm ² . We also saw that $52^2 - 48^2 = 4 \times 100 = 400$ matches $2704 - 2304 =$	The identity $a^2 - b^2 = (a - b)(a + b)$ is true for all values of a and b. It gives a quick way to calculate differences of squares mentally. Together with $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$, it makes up the three identities in our model. The plot area $x(20 - x)$ can be written as $100 - (x -$	Cutting the L-shape and rearranging the pieces does not change the area, so $a^2 - b^2$ must equal $(a - b)(a + b)$. This helps explain Mama Wanjiru's plot in Kericho: the area is 100 minus a small square, which hints at why widths on either side of 10 m give equal areas (8 m and 12 m are the

	400, and that for the Kericho plot, $x = 8$ gives $100 - 4 = 96 = 8 \times 12$.	$10)^2$, a preview we will use fully in Lesson 8.	same 8 m by 12 m plot turned round) and why 10 m gives the most, 100 m ² . We still need to learn how to factorise and solve to find widths for a target area without trial and error.
Lesson 4 Factorising $x^2 + bx + c$: Running the Expansion Backwards to Find the Plot	We built rectangles from x^2 tiles, x strips and unit squares. $x^2 + 8x + 15$ made a rectangle with sides $(x + 3)$ and $(x + 5)$. For $x^2 - 20x + 96$ we listed the factor pairs of 96 and found that $8 \times 12 = 96$ and $8 + 12 = 20$.	To factorise $x^2 + bx + c$, find two numbers p and q whose sum is b and whose product is c , then write $(x + p)(x + q)$. Check by expanding. When the sum is negative and the product is positive, both numbers are negative.	$x^2 - 20x + 96 = (x - 8)(x - 12)$ because $-8 + -12 = -20$ and $(-8)(-12) = 96$. The numbers 8 and 12 are the widths that gave 96 m ² on Mama Wanjiru's 40 m fence in Kericho. They add up to 20 m, so the 8 m and 12 m plots are the same 8 m by 12 m rectangle turned round.
Lesson 5 Factorising $ax^2 + bx + c$ ($a \neq 1$) and Special Cases: Which Method Works Best?	We built $2x^2 + 7x + 3$ from area tiles (two x^2 tiles, seven x strips, three unit squares) and found a rectangle with sides $(2x + 1)$ and $(x + 3)$. We also saw that $x^2 + 10x + 25$ makes a square and that $x^2 - 100$ has no middle term.	To factorise $ax^2 + bx + c$ with $a \neq 1$, find two numbers with product ac and sum b , split the middle term, group, take out common factors, and check by expansion. Perfect squares and differences of two squares are special cases to look for first.	Factorising turns an area back into the sides of a rectangle, just as $x^2 - 20x + 96 = (x - 8)(x - 12)$ gives the widths 8 m and 12 m for Mama Wanjiru's 96 m ² plot. Which method is best depends on the expression: a special case, sum and product when $a = 1$, or grouping or tiles when $a \neq 1$. We still need to use the factors to find x , which is the next lesson.
Lesson 6 Two Widths, One Area: Solving Quadratic Equations by Factorisation	We saw that $x(20 - x) = 96$ can be rearranged to $x^2 - 20x + 96 = 0$ and factorised as $(x - 8)(x - 12) = 0$. Substituting $x = 8$ and $x = 12$ into $x(20 - x)$ both gave 96 m ² , while $x = 9$ did not make the product zero.	We learned that a quadratic equation is solved by rearranging to zero, factorising and using the zero-product property: if $p \times q = 0$ then $p = 0$ or $q = 0$. We also learned to check each solution against the context $0 < x < 20$.	The two solutions $x = 8$ and $x = 12$ appear because width + length = 20 m, so if the width is 8 m the length is 12 m and vice versa. They describe the same 8 m by 12 m plot turned round, not two different shapes, which resolves the puzzle in Mama Wanjiru's plot. Our model now includes two solutions that add up to 20.
Lesson 7 Forming and Solving Quadratic Equations from Real-Life Situations: Mandazi Tray, Path Around the Shamba and Thrown Ball	We observed that a worded situation (60 mandazi in x rows and $(x + 4)$ columns; a path of width w around a 12 m by 8 m shamba with total area 140 m ² ; a ball with $h = 20t - 5t^2$) can be written as an equation and rearranged to zero. We saw that the ball reaches 15 m twice ($t = 1$ and $t = 3$), like the 96 m ² area appearing at $x = 8$ and $x = 12$ on the 40 m fence.	We learned the steps: define the variable with units, write the equation, rearrange to $ax^2 + bx + c = 0$, factorise, use the zero-product property, then check each root in the situation. The mandazi equation $x^2 + 4x - 60 = 0$ gives $(x + 10)(x - 6) = 0$; the path equation $w^2 + 10w - 11 = 0$ gives $(w + 11)(w - 1) = 0$; the ball equation $t^2 - 4t + 3 = 0$ gives $(t - 1)(t - 3) = 0$.	We can explain that a quadratic equation can have two roots but the context decides which ones are answers: $x = -10$ rows and $w = -11$ m are rejected because lengths cannot be negative, so the answers are $x = 6$ and $w = 1$, while $t = 1$ s and $t = 3$ s are both kept because the ball goes up and comes down. This is why Mama Wanjiru's plot has two valid widths ($x = 8$ and $x = 12$, the same 8 m by 12 m plot turned round) but widths above 20 m are impossible.
Lesson 8 Applying and Synthesising: Design a	We observed that with 40 m of wire the length is $20 - x$, so the area is $x(20 - x)$. A target of 84 m ² gives $x^2 - 20x + 84 = 0$,	We learned how to form a quadratic equation from a target area, rearrange it to zero, factorise it, use the zero-product	Two widths such as 8 m and 12 m, or 6 m and 14 m, give the same area because x and $20 - x$ describe the same rectangle

School-Garden Plot and Explain the Maximum	which gives a 6 m by 14 m plot. A target of 105 m ² has no valid width, and the area table never goes above 100 m ² , reached at $x = 10$.	property, and check the solutions against $0 < x < 20$. We learned that $A = 100 - (x - 10)^2$ shows the area can never exceed 100 m ² .	turned round, so the area is symmetric about $x = 10$. There is a maximum of 100 m ² at $x = 10$ because a square $(x - 10)^2$ cannot be less than 0, so no plot with 40 m of wire can enclose more than 100 m ² .
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ATTRIBUTION AND LICENSING

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